

FINGERPRINT SENSOR

Technology, Working Principle & AI/ML

Technical Seminar • Artificial Intelligence and Expert Systems (MLA01)

SIMATS Engineering

Presented by: Akhil Arthimalla

Reg. No.: 192525283

1. Fingerprint Sensor: Technology & Working

A sensor converts fingerprint ridge/valley patterns into digital data.

Optical

Uses reflected light to create a fingerprint image.

Capacitive

Measures capacitance changes caused by ridges and valleys.

Ultrasonic

Uses acoustic echoes; supports 3-D and in-display sensing.

Basic architecture



Capture → digitize → enhance → extract features → compare with an enrolled template.

Key parameters: accuracy, resolution, power, response time, cost and spoof resistance.

2. AI/ML Integration & Applications

AI adds intelligence to sensing, quality assessment and security.

AI / ML

- Quality assessment
- Liveness / spoof detection
- Deep feature extraction

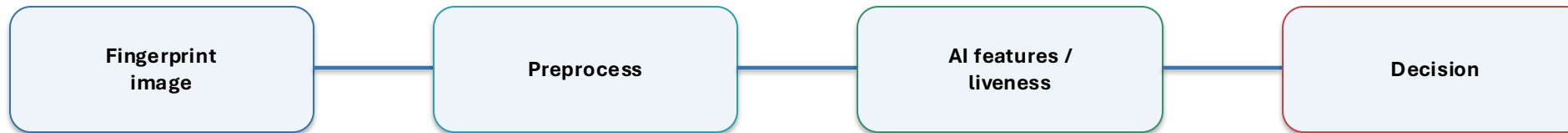
Applications

- Smartphones
- Access control
- Identity systems
- Forensics

Example

Ultrasonic in-display fingerprint sensors combine advanced sensing with secure biometric processing.

AI-enabled pipeline



AI goal: improve accuracy and detect presentation attacks without excessive latency or power.

Recent research focuses on deep-learning liveness detection and contactless fingerprint recognition.

3. Comparison, Challenges & Future

Key points for technical defense

Optical

Mature imaging; simple concept; optical path affects performance.

Capacitive

Compact and fast; requires careful electrode and analog design.

Ultrasonic

3-D information and strong in-display potential; more complex.

Challenges

Spoofing • wet/dry fingers • sensor variation • privacy • template security

Future trends

Contactless sensing • larger ultrasonic areas • on-device AI • multimodal biometrics

Main idea

The best sensor depends on the required balance of accuracy, cost, power, integration and security.

4. Conclusion

Key takeaway

Fingerprint sensors are evolving from simple scanners into intelligent biometric systems.

Sensor technology + Signal processing + AI/ML + Security

Today

Fast and convenient biometric authentication.

AI advantage

Better quality assessment and spoof detection.

Future

Contactless, low-power and privacy-focused systems.

Overall: sensor choice should balance accuracy, cost, power, usability and security.